

Grace — PRIMARY 3: the 0.036 reconciliation — T2001 and F489 unify as leading-vs-full of ONE curvature correction

2026-07-13 Monday. Keeper PRIMARY 3 (my self-contained lane): reconcile the two corpus readings of the 0.036 that turns $\alpha^{-1}=137$ into 137.036 — T2001 (“running”: $\alpha^{-1}=137+n_C/N_max$) vs F489 (Wyler: the exact 137.036 is already in V_Wyler). Casey’s reading: $\kappa_Bergman=-n_C$, so $n_C/N_max = \text{curvature/capacity}$ (the two-sided ellipse). RESULT: they unify — n_C/N_max is the leading curvature term of the exact Wyler correction, and the curvature reading rests on a PINNED invariant. The exact same-measure identification stays F489’s open gate; α stays IDENTIFIED.

The three readings of $\delta = \alpha^{-1} - 137$

reading	α^{-1}	δ	residual vs obs
observed (CODATA)	137.035999	0.035999	—
V_Wyler (F489): ($9/8\pi^4$)($\pi^5/2^4 \cdot 5!$) ^{1/4}	137.036082	0.036082	+0.23% of δ
$137 + n_C/N_max$ (T2001)	137.036496	0.036496	+1.38% of δ

$n_C/N_max \div (V_Wyler-137) = 1.0115$ — so n_C/N_max is the leading term of the exact Wyler correction (they agree to 1.1%; the higher-order curvature terms in V_Wyler bring it down toward observed). T2001 $\subset$ F489: the same curvature correction, read at leading (discrete) order vs full (geometric) order. They were never competing.

The curvature reading is GROUNDED on a pinned invariant (not an assertion)

Casey: $n_C/N_max = |\kappa_Bergman|/N_max = \text{curvature/capacity}$. And $\kappa_Bergman = -n_C$ is a PINNED BST result (Helgason 1962, Toy 3661; registered) — the Bergman/holomorphic curvature of D_IV^5 . (Companion: $Ric = -(n_C+1)g_B = -C_2 \cdot g_B$, Kähler-Einstein.) So the leading correction is literally the domain’s curvature over its capacity: - capacity = $N_max = 137 = N_c^3 \cdot n_C + \text{rank}$ (the discrete mode-count, T1939 — this morning’s α -keystone base); - curvature/capacity = $|\kappa_Bergman|/N_max = n_C/N_max$ (the continuous correction the boundary carries).

This realizes Casey’s discrete/continuous principle exactly: $\alpha^{-1} = (\text{discrete mode-count}) + (\text{continuous curvature correction})$. In the ellipse form $\alpha^{-1} = a + \kappa/a$ with $a = N_max = 137$, $\kappa = |\kappa_Bergman| = n_C = 5 \rightarrow 137 + 5/137$ (=T2001).

What this closes, and what stays open (α stays IDENTIFIED)

- CLOSED (reconciliation): T2001 and F489 are unified — one curvature correction, leading (n_C/N_max) vs full (V_Wyler). The numerator n_C is the pinned Bergman curvature $|\kappa_Bergman|$, not a free fit. The structure (capacity + curvature/capacity) is confirmed and grounded. This resolves the “two competing readings” flag.
- OPEN (F489’s bank gate, NOT closed): the *exact* value needs the full boundary-curvature integral. n_C/N_max is only the leading term (1.4%); V_Wyler is fuller (0.2%); neither is exact — both approximate the true integral. And the load-bearing identification — is V_Wyler literally the $S^2 \times S^1$ coupling-curvature integral (same measure)? — is F489’s ellipse-vs-epicycle test, still open. I did NOT analytically expand V_Wyler to show n_C/N_max drops out as its leading term (that expansion is the remaining rigor); the reconciliation here is numerical + curvature- grounded, not a proven expansion.
- So α stays IDENTIFIED (Wyler), not derived. This reconciliation sharpens the *structure* of the 0.036 (it’s a genuine curvature effect with a pinned numerator) but does not close the same-measure identification.

Honest tier

- Reconciliation T2001 $\subset$ F489 (leading vs full): SOLID — arithmetic + the pinned $\kappa_{\text{Bergman}} = -n_C$.
- $n_C/N_{\text{max}} = |\kappa_{\text{Bergman}}|/N_{\text{max}} = \text{curvature/capacity}$: GROUNDED reading (pinned invariant), numerically consistent at leading order (1.1%). NOT a derivation of the exact value.
- V_{Wyler} = the coupling-curvature integral (same measure): OPEN — F489's bank gate. Cal #27 respected: I do NOT claim the 0.036 is derived; it's a curvature-structured correction, leading term found, exact integral open.
- No bank, graph metadata updated (SOD current, max T2517).

— Grace, 2026-07-13. PRIMARY 3: the 0.036 is one curvature correction read two ways — T2001 (n_C/N_{max} , leading, discrete) and F489 (V_{Wyler} , full, geometric) unify (agree to 1.1%), with the numerator = the pinned Bergman curvature $|\kappa_{\text{Bergman}}| = n_C$ (Helgason 1962). Capacity (137, mode-count) + curvature/capacity (n_C/N_{max}) = Casey's discrete/continuous ellipse. Exact value + the V_{Wyler} =coupling-curvature same-measure identification stay open (F489's gate); α stays IDENTIFIED.